Node class	num of nodes	cap. per node	activity factor
local logic outputs	20000	3	0.12
medium wires	4000	18	0.08
long global wires	600	95	0.03

I apologise for the trouble! Due to my disability, my wrists and hands are not working well at the moment so I must type all my work. I hope that I have formatted my work in a clear enough manner to mark, but if not, please let me know and I will explain my reasonings over a Zoom call or in person.



Supply V	0.9	
Clock Freq	1.2	Ghz

1A

Pdyn = sum(Ci*V^2*f*activity factor)

Node class	num of nodes	cap. per node	Supply V	f	activity factor	**Pdyn**
local logic outputs	20000	3	0.9	1.2	0.12	**6998.4**
medium wires	4000	18	0.9	1.2	0.08	**5598.72**
long global wires	600	95	0.9	1.2	0.03	**1662.12**

The local logic output class dominates.

The answer is not obvious from capacitance alone; I would have guessed that the long global wires would have had the highest Dynamic Power, because they had the highest capacitance per node and Pdyn is proportional to the capacitance per node, but the other factors were large enough that the answer was not obvious simply from the capacitance values.

1B

Total dynamic switching power: sum of all dynamic power in circuit.

local logic outputs	6998.4
medium wires	5598.72
long global wires	1662.12

sum = 14259.24

Total dynamic switching power: 14259.24 equals 14.3mW

total short circuit power: **8% of total dynamic switching power.**

total dynamic switching power= 14259.24

8% total dynamic switching power= 1140.7392

total short circuit power: **1140.7392** 11.4mW

1C

leakage current per minimum-equivalent transistor 2.5 nA

num of transistors 1800000

Total leakage current: leakage per transistor * num of transistors 4500000

Power = Current * Voltage, P=IV V=0.9

Total leakage power = **4050000** **equals** **4.05 mW**

Dynamic switching power: 14259.24 equals 14.3mW

(from previous question)

Total leakage power is far lower than the dynamic switching power.

1D

Old design:

Node class	num of nodes	cap. per node	activity factor
local logic outputs	20000	3	0.12
medium wires	4000	18	0.08
long global wires	600	95	0.03
Supply V	0.9		
Clock Freq	1.2	Ghz	

Pdyn = sum(Ci*V^2*f*activity factor)

Node class	num of nodes	cap. per node	Supply V	f	activity factor	**Pdyn**
local logic outputs	20000	3	0.9	1.2	0.12	**6998.4**
medium wires	4000	18	0.9	1.2	0.08	**5598.72**
long global wires	600	95	0.9	1.2	0.03	**1662.12**
					sum Pdyn:	14259.24

New design:

Node class	num of nodes	cap. per node	activity factor
local logic outputs 1	20000	3	0.12
local logic outputs 2	5000	3	0.05
medium wires	4000	18	0.08
long global wires	600	95	0.03

Node class	num of nodes	cap. per node	Supply V	f	activity factor	**Pdyn**
local logic outputs 1	20000	3	0.9	1.2	0.12	**6998.4**
local logic outputs 2	5000	3	0.9	1.2	0.05	**729**
medium wires	4000	18	0.9	1.2	0.08	**5598.72**
long global wires	600	95	0.9	1.2	0.015	**831.06**
					sum Pdyn:	14157.18

Yes, this change would reduce total dynamic power as shown above. The old Pdyn total was 14259, and the new value is 14157.

1E

The student is wrong, but I cannot prove them wrong with my numerical results. At all points, the local logic output node class use the most dynamic power in total, but if the other factors (capacitance, activity factor, etc. were different, then there is no reason why another node class couldn't dominate.